



NTI DATA ANALYSIS FINAL PROJECT

RAW DATA TO ACTIONABLE INSIGHTS

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OUR TEAM

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Exploring The Data

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Data Processing And Cleaning

03 **Mahmoud Hassan**
Answering Bussiness Questions

04 **Youssef Aly**
Dashboard And Reporting

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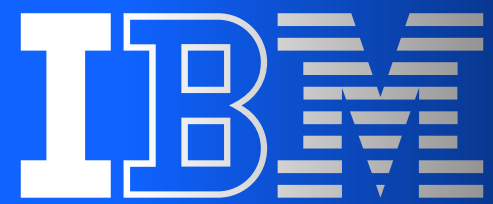
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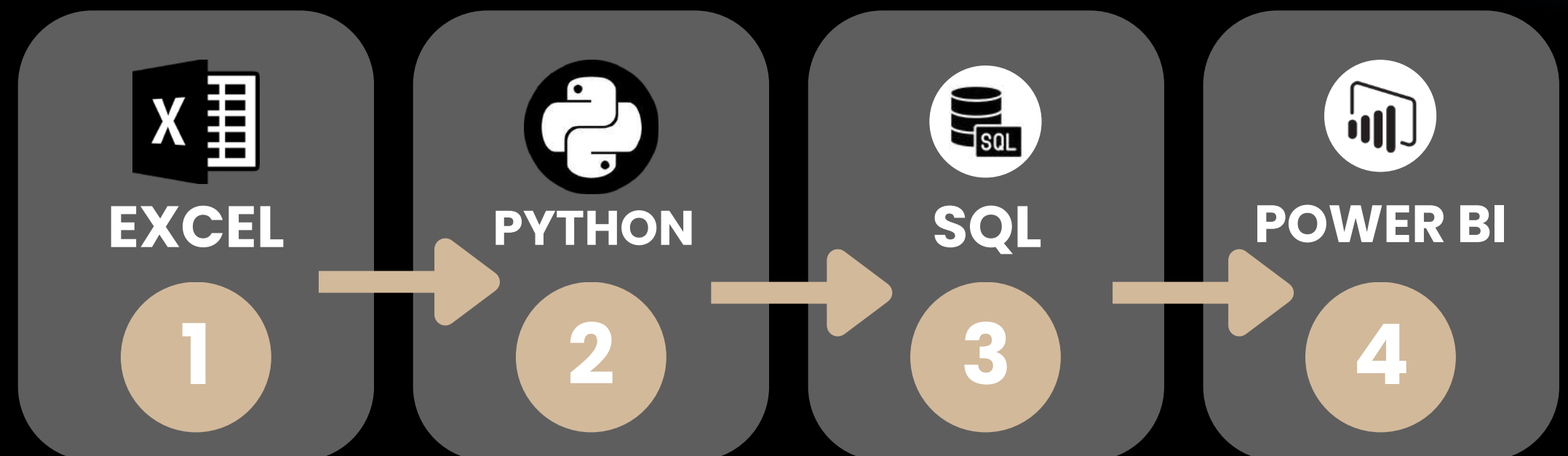


01

BUSINESS PROBLEM & OBJECTIVES

Our Team act as People Analytics analysts for a IBM's HR department, investigating why employees leave and which departments/roles are most affected, to inform retention strategy.

WORKFLOW



01

BUSINESS PROBLEM & OBJECTIVES



Exploring & Cleaning The Data

Understanding the data we are dealing with, exploring dataset inconsistencies, null values, duplicate data, and the kinds of columns we are dealing with, and then cleaning it.



Answering Business Questions

After Cleaning the Dataset, We go on exploring what the data have to offer by answering business questions, discovering patterns and extracting important insights.



Dashboarding & Reporting

Now that we have explored the dataset and extracted important insights, we can move on to visualizing the data for non-technical people, making the insights easier to understand and communicate.

02 ABOUT THE DATA

Dataset Name

- IBM HR Analytics Employee Attrition & Performance

Dataset Source

- <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Dataset Format

- CSV



02

ABOUT THE DATA (DATASET AT A GLANCE)

1470

Employee Records

35

Columns

3

One Value Columns

CORE METRICS / COLUMNS

Attrition | • YES
• NO

Depart. | • SALES
• HUMAN REOURCES
• RESEARCH & DEVELOPMENT

Marital
Statues | • SINGLE
• MARRIED
• DIVORCED

Job Role

- HEALTHCARE REPRESENT.
- HUMAN RESOURCES
- LABORATORY TECHNICIAN
- MANAGER
- MANUFACTURING DIRECTOR
- RESEARCH DIRECTOR
- RESEARCH SCIENTIST
- SALES EXECUTIVE
- SALES REPRESENTATIVE

EmployeeCount | • 1 ONLY

Over 18 | • Y ONLY

StandardHours | • 80 ONLY

02

ABOUT THE DATA (DATASET AT A GLANCE)

Important KPIs

Total Employees

1,470

Attrition Count

237

Attrition Rate

16.12%

Avg. Monthly Income

6502.93

Avg. Job Satisfaction

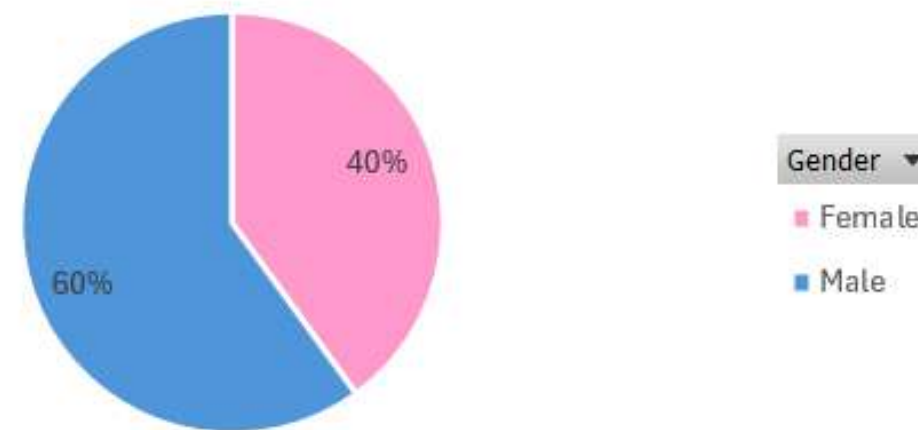
2.73

1

How Are Genders Distributed Across the Dataset?

Count of EmployeeNumber

Gender Distribution

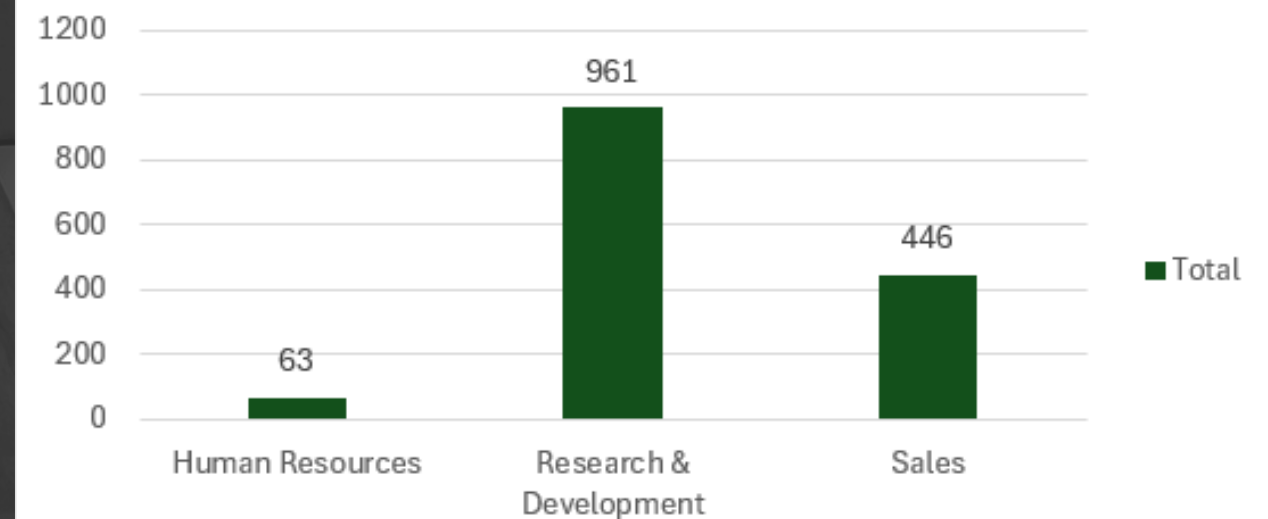


2

How Are Employees Distributed Across Departments?

Count of EmployeeNumber

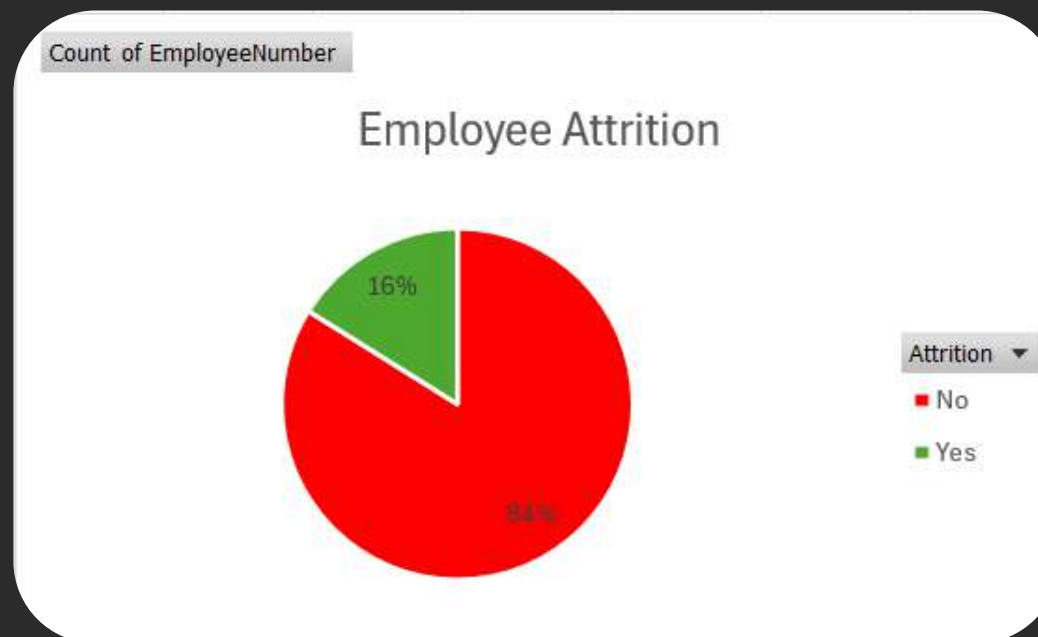
Employees by Department



02 ABOUT THE DATA (DATASET AT A GLANCE)

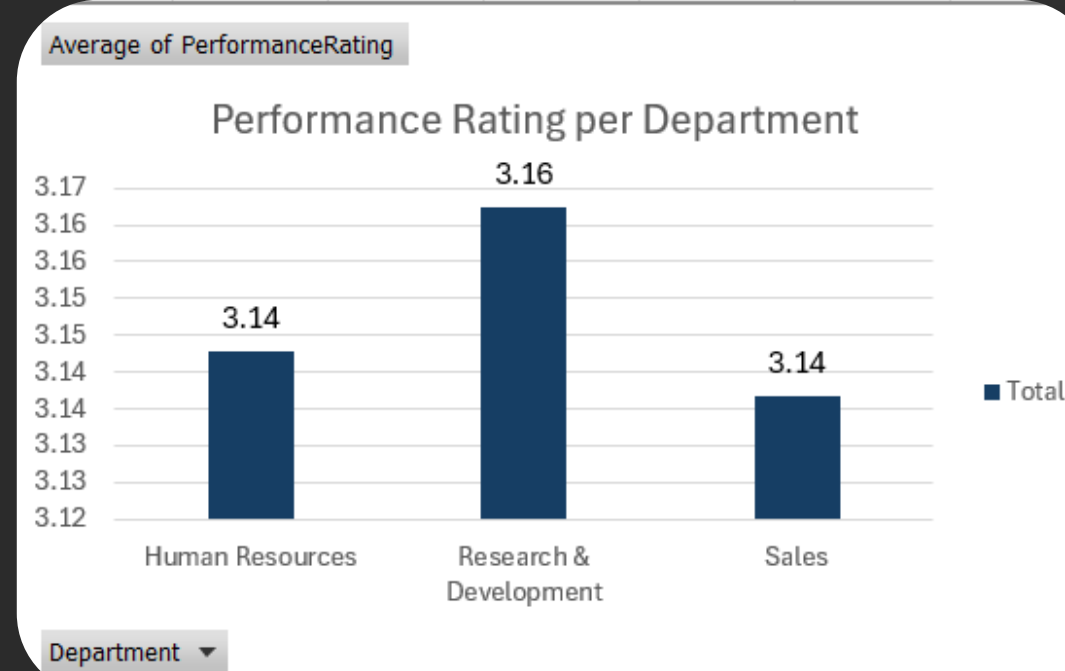
3

What Is the Attrition Rate Distribution?



4

What Is the Performance Rating Per Department??



5

What is the distribution of job satisfaction levels among employees?



An illustration on the left side of the slide shows three blue database cylinders. A yellow cleaning brush with a brown handle is positioned as if cleaning the cylinders. The background is dark blue with a subtle gradient.

03

DATA CLEANING AND PROCESSING

1

No Duplicates Detected

2

No Nulls Found



```
print(f"Dataset Shape: {dataset.shape}")  
print(f"Duplicates Count: {dataset.duplicated().sum()}")  
print(f"Missing Values Total: {dataset.isna().sum().sum()}")
```



```
Dataset Shape: (1470, 35)  
Duplicates Count: 0  
Missing Values Total: 0
```


An illustration on the left side of the slide shows three blue database cylinders. A yellow torch with a brown handle is positioned in front of them, pointing towards the right. The background is a dark blue gradient.

03

DATA CLEANING AND PROCESSING

3

Dropped 4 Columns
*All containing One Value



```
constant_cols = [col for col in dataset.columns if
dataset[col].nunique() <= 1]
cols_to_drop = constant_cols + ["EmployeeNumber"]
print(f"Columns to drop: {cols_to_drop}")

# Output
Columns to drop:
['EmployeeCount', 'Over18', 'StandardHours', 'EmployeeNumber']
```

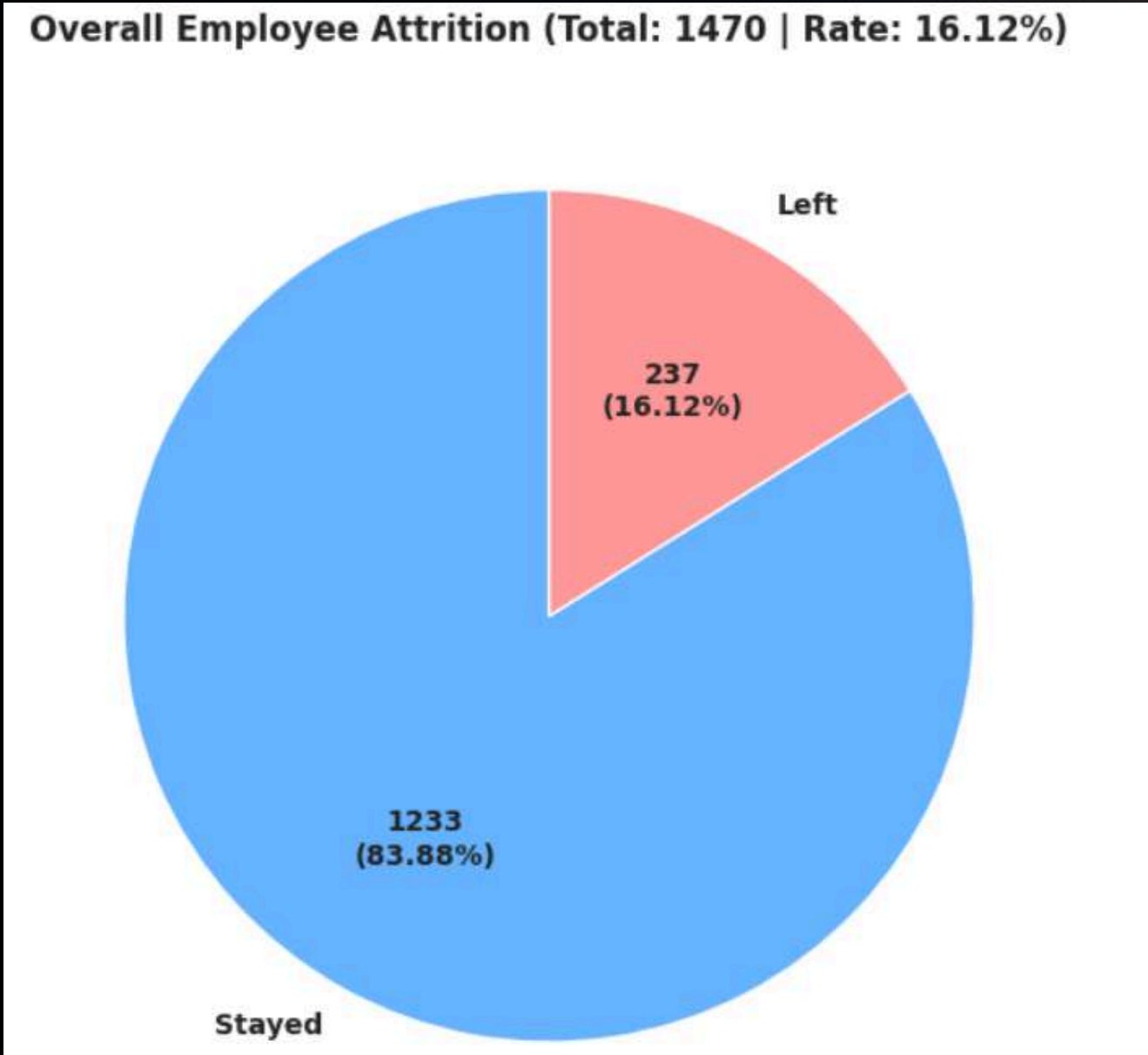
04 EXPLORATORY ANALYSIS

(ANSWERING BUSINESS QUESTIONS)

Understand overall attrition rate and how it varies across departments

```
select
  count(*) as Total_Emps,
  sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,
  sum(case when Attrition = 0 then 1 else 0 end) as Emps_Stayed,
  cast(
    sum(case when Attrition = 1 then 1 else 0 end) * 100.0
    / count(*)
    as decimal(5,2)
  ) as Attrition_Rate
from Employees;
```

	Total_Emps	Emps_Left	Emps_Stayed	Attrition_Rate
1	1470	237	1233	16.12



04

EXPLORATORY ANALYSIS

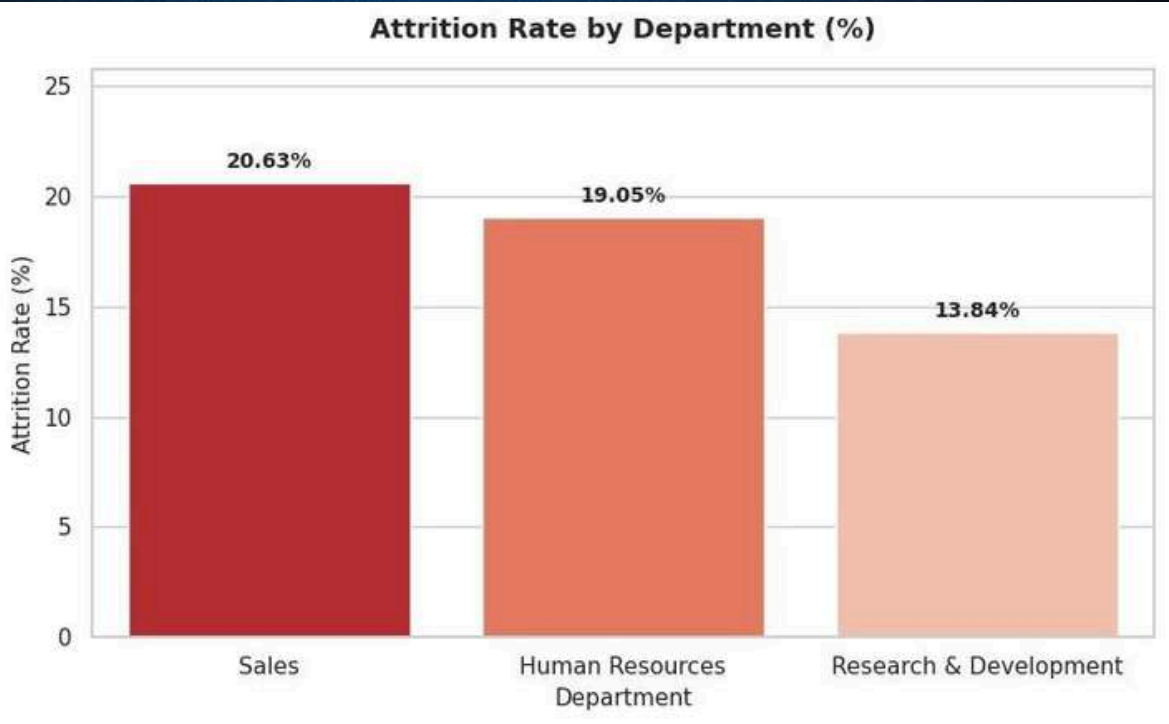
(ANSWERING BUSINESS QUESTIONS)

What is the attrition rate by department, job role, and overtime status?

DEPARTMENT

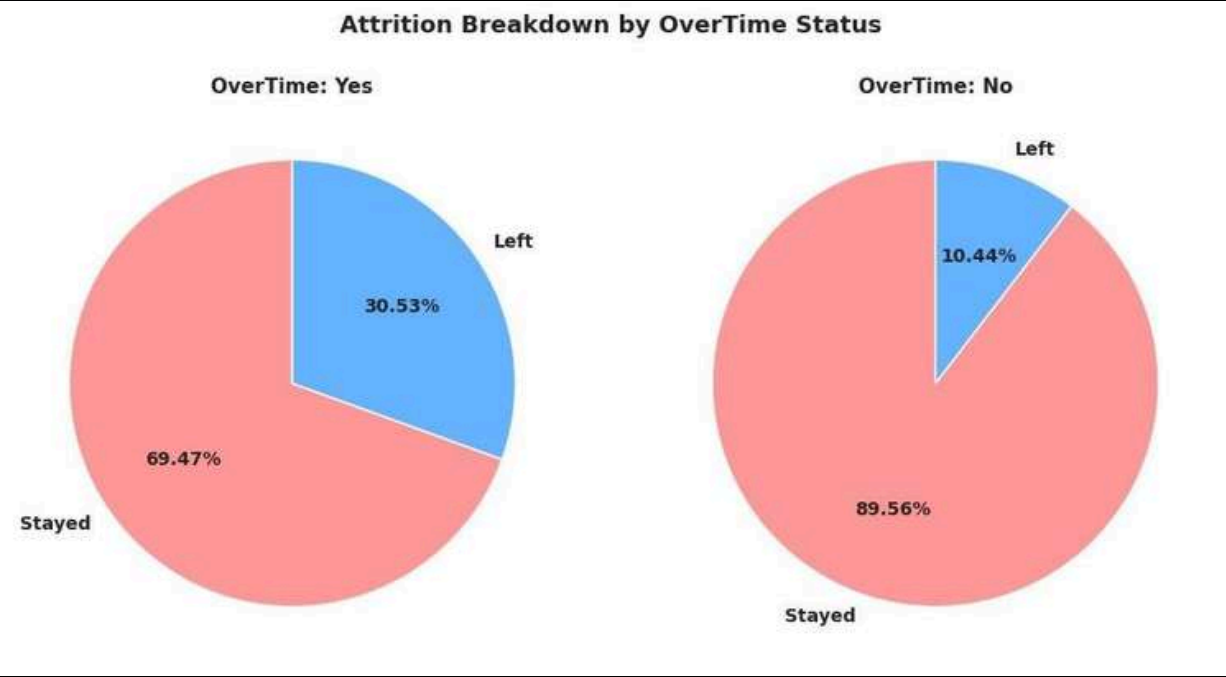
```
select
  Department,
  count(*) as Total_Emps,
  sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,

  cast(
    sum(case when Attrition = 1 then 1 else 0 end) * 100.0
    / count(*)
    as decimal(5,2)
  ) as Attrition_Rate
from Employees
group by Department
order by Attrition_Rate desc;
```



OVERTIME

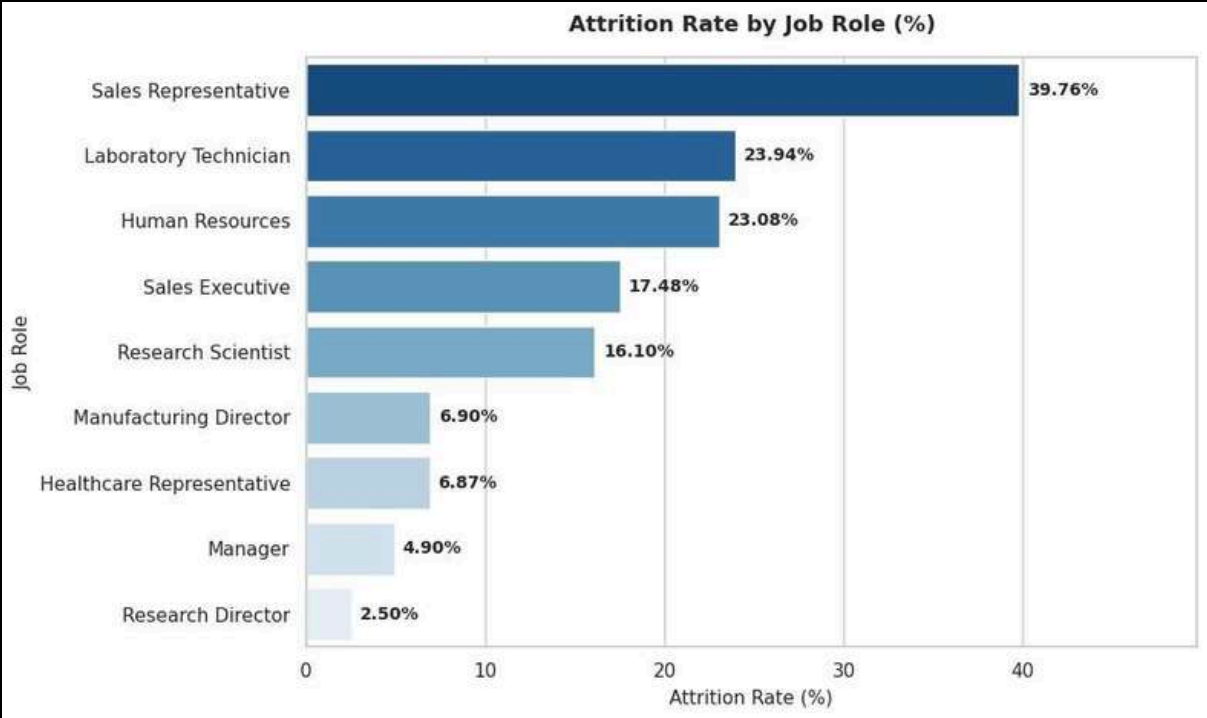
```
select
  case
    when OverTime = 1 then 'Yes'
    else 'No'
  end as OverTime_Status,
  count(*) as Total_Emps,
  sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,
  cast(sum(case when Attrition = 1 then 1 else 0 end) * 100.0 /
  count(*) as decimal(5,2)
  ) as Attrition_Rate
from Employees
group by OverTime
order by Attrition_Rate desc;
```



JOB ROLE

```
select
  JobRole,
  count(*) as Total_Emps,
  sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,

  cast(
    sum(case when Attrition = 1 then 1 else 0 end) * 100.0
    / count(*)
    as decimal(5,2)
  ) as Attrition_Rate
from Employees
group by JobRole
order by Attrition_Rate desc;
```



04

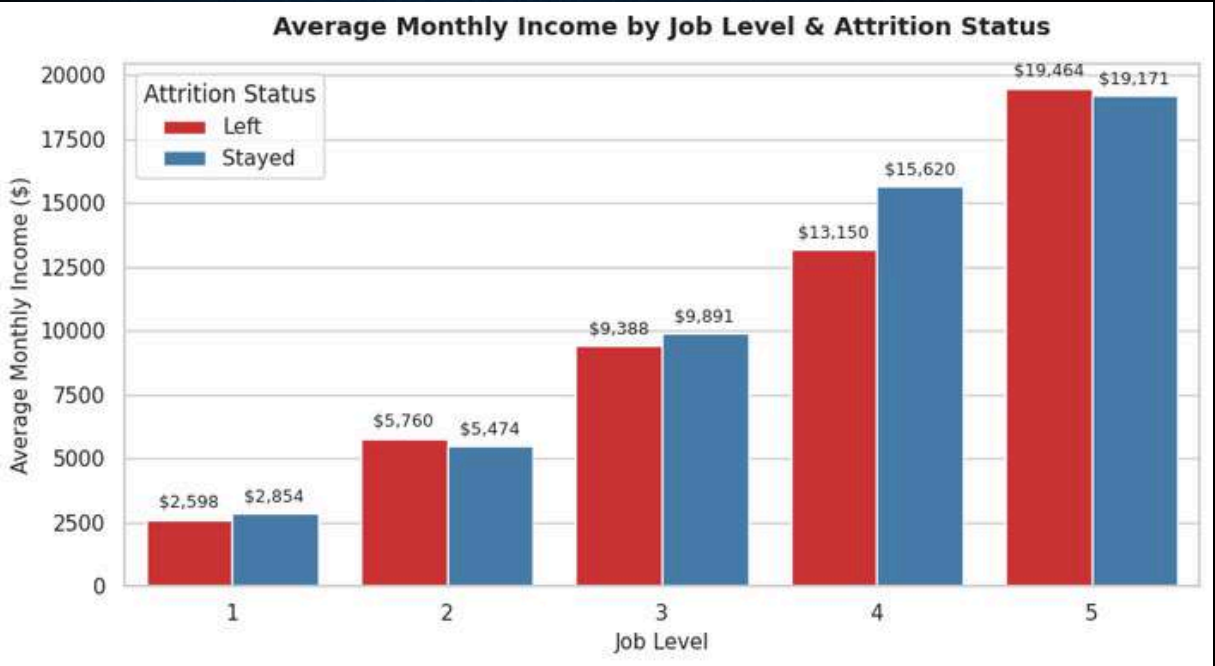
EXPLORATORY ANALYSIS

(ANSWERING BUSINESS QUESTIONS)

What is the average monthly income by job level and attrition status?

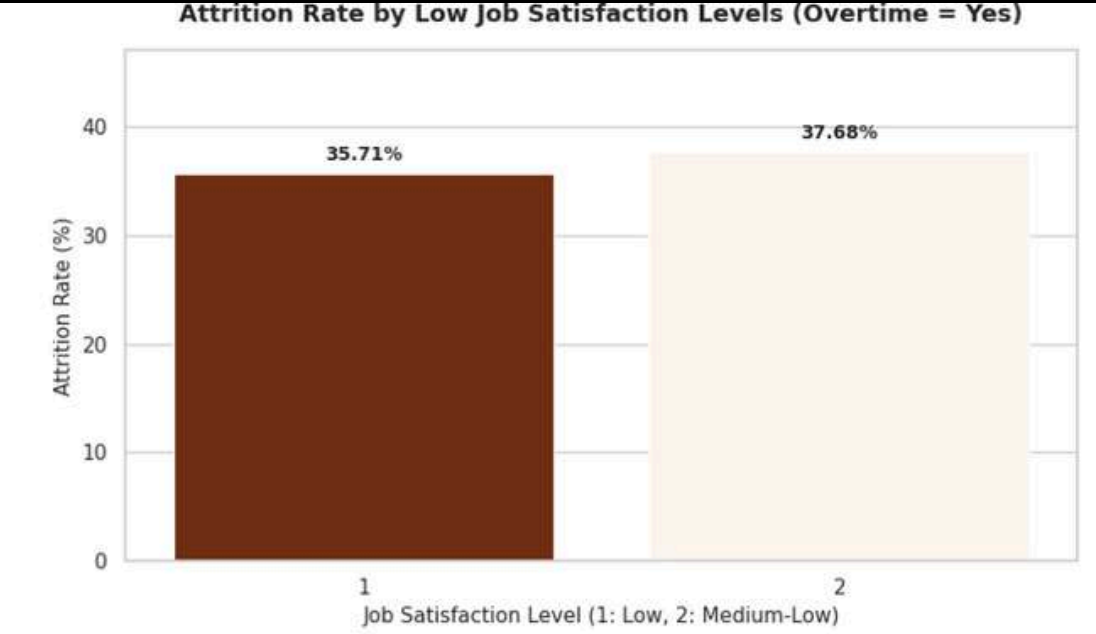
```
select
  JobLevel,
  case
    when Attrition = 1 then 'Left'
    else 'Stayed'
  end as AttritionStatus,

  cast(avg(MonthlyIncome) as decimal(10,2)) as Avg_MonthlyIncome
from Employees
group by
  JobLevel,
  Attrition
order by
  JobLevel,
  AttritionStatus;
```



Which employees combine low job satisfaction with frequent overtime?

```
select
  JobSatisfaction,
  case
    when OverTime = 1 then 'Yes'
    else 'No'
  end as OverTimeStatus,
  count(*) as Total_Emps,
  sum(case when Attrition = 1 then 1 else 0 end) as Emps_left,
  cast(
    sum(case when Attrition = 1 then 1 else 0 end) * 100.0
    / count(*)
  ) as AttritionRate
from Employees
where JobSatisfaction <= 2
and OverTime = 1
group by
  JobSatisfaction,
  OverTime
order by
  JobSatisfaction;
```

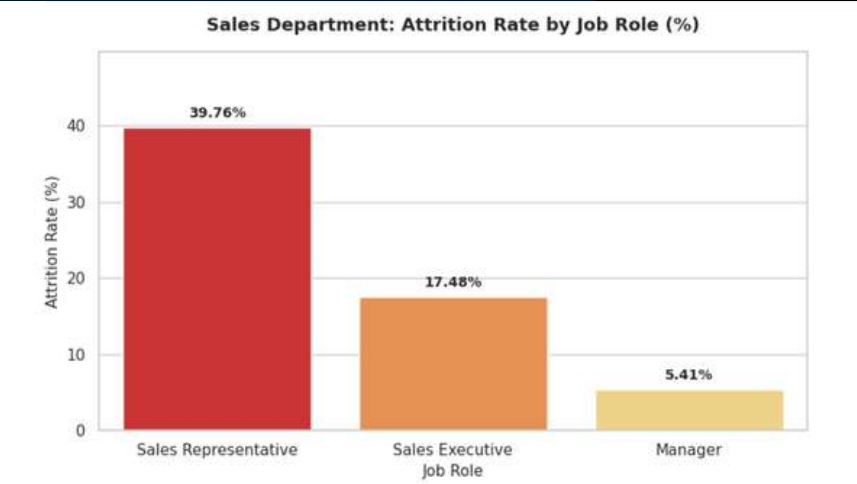


Which department has the highest attrition, and how does it break down by job role within that department?
+ Diagnostic

```
select top 1
  Department,
  cast(
    sum(case when Attrition = 1 then 1 else 0 end) * 100
    / count(*)
    as decimal(5,2)
  ) as AttritionRate
from Employees
group by Department
order by AttritionRate desc;
```

	Department	AttritionRate
1	Sales	20.63

```
select
  JobRole,
  cast(
    sum(case when Attrition = 1 then 1 else 0 end) * 100.0
    / count(*)
    as decimal(5,2)
  ) as AttritionRate
from Employees
where Department = 'Sales'
group by JobRole
order by AttritionRate desc;
```



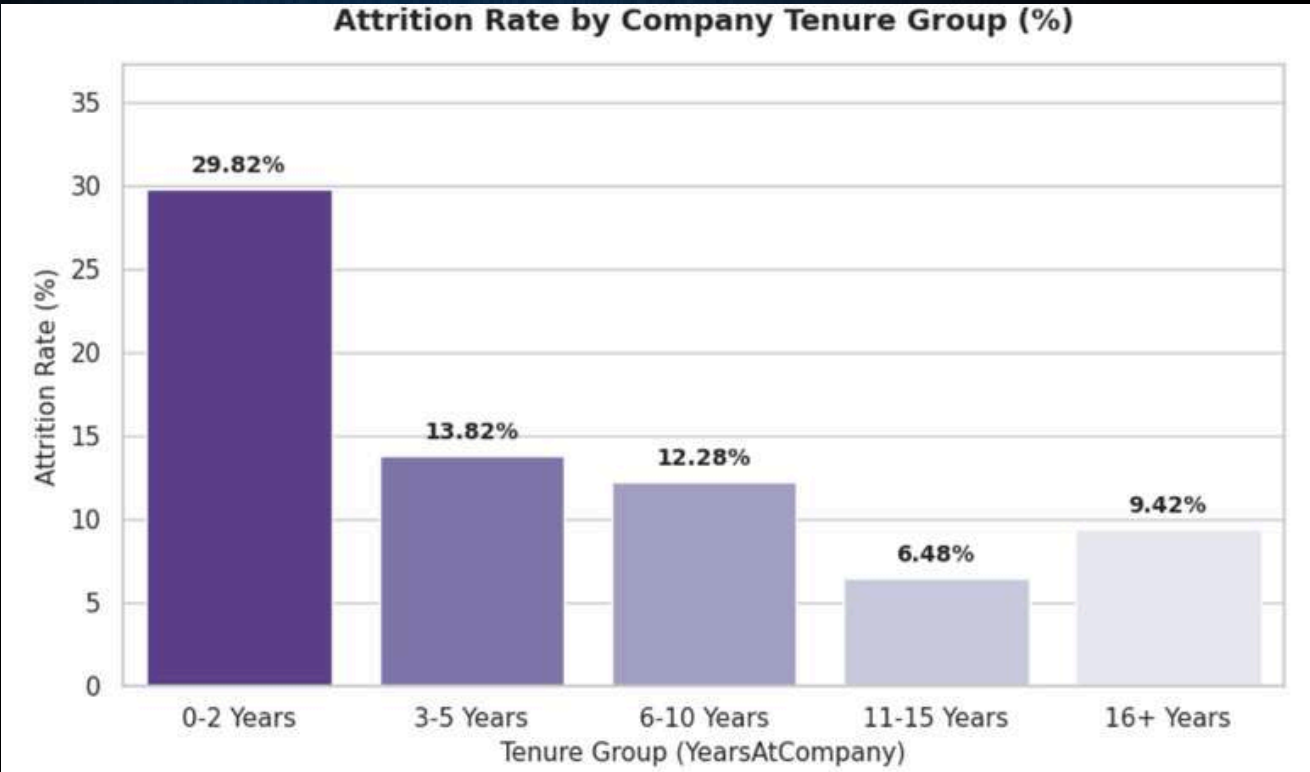
04

EXPLORATORY ANALYSIS

(DIAGNOSTIC ANALYTICS QUESTIONS)

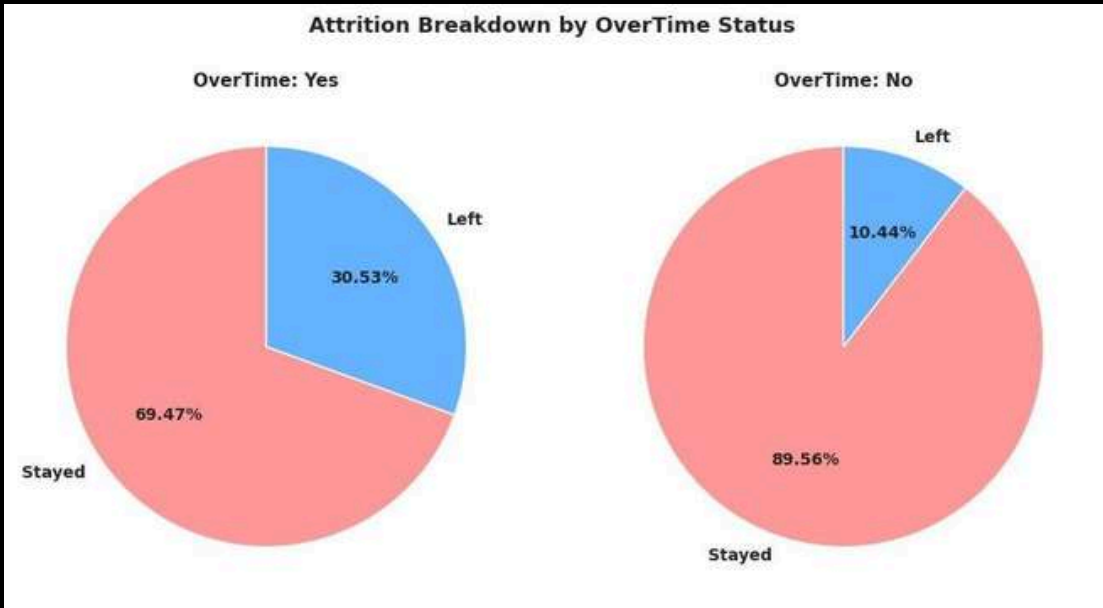
Bucket "years at company" into tenure groups and compute attrition rate per group

```
select
  case
    when YearsAtCompany <= 2 then '0-2 Years'
    when YearsAtCompany <= 5 then '3-5 Years'
    when YearsAtCompany <= 10 then '6-10 Years'
    when YearsAtCompany <= 15 then '11-15 Years'
    else '16+ Years'
  end as TenureGroup,
  cast(
    sum(case when Attrition = 1 then 1 else 0 end) * 100.0
    / count(*)
    as decimal(5,2)
  ) as AttritionRate
from Employees
group by
  case
    when YearsAtCompany <= 2 then '0-2 Years'
    when YearsAtCompany <= 5 then '3-5 Years'
    when YearsAtCompany <= 10 then '6-10 Years'
    when YearsAtCompany <= 15 then '11-15 Years'
    else '16+ Years'
  end;
```



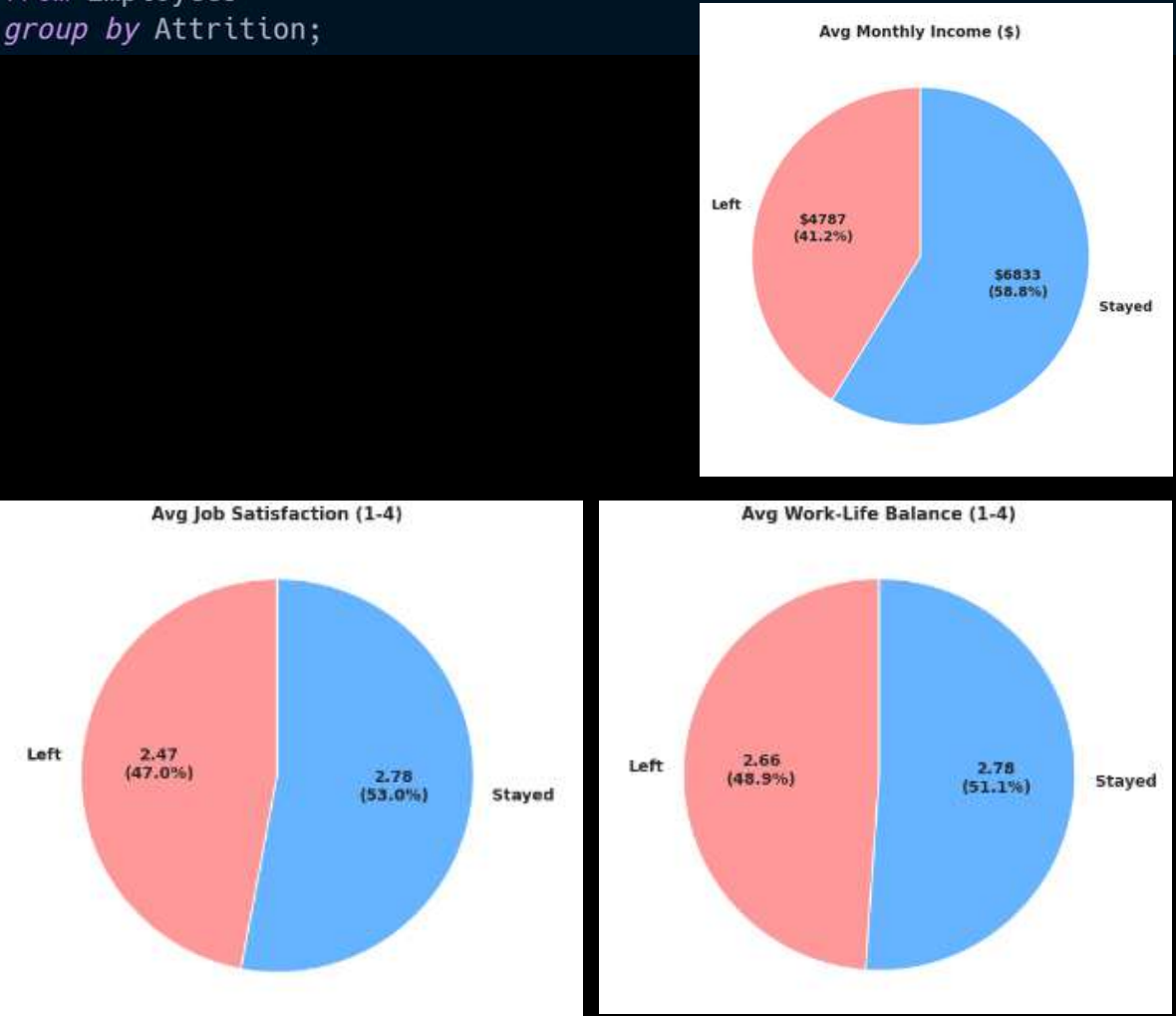
Cross-tabulate overtime status against attrition, and job satisfaction score against attrition

```
select
  case
    when OverTime = 1 then 'Yes'
    else 'No'
  end as OverTime_Status,
  sum(case when Attrition = 1 then 1 else 0 end) as Emps_Left,
  sum(case when Attrition = 0 then 1 else 0 end) as Emps_Stayed
from Employees
group by OverTime;
```



Compare average income, satisfaction, and work-life balance scores between employees who left vs. stayed

```
select
  case
    when Attrition = 1 then 'Left'
    else 'Stayed'
  end as AttritionStatus,
  avg(MonthlyIncome) as Avg_Income,
  avg(JobSatisfaction) as Avg_Satisfaction,
  avg(WorkLifeBalance) as Avg_WorkLife_Balance
from Employees
group by Attrition;
```



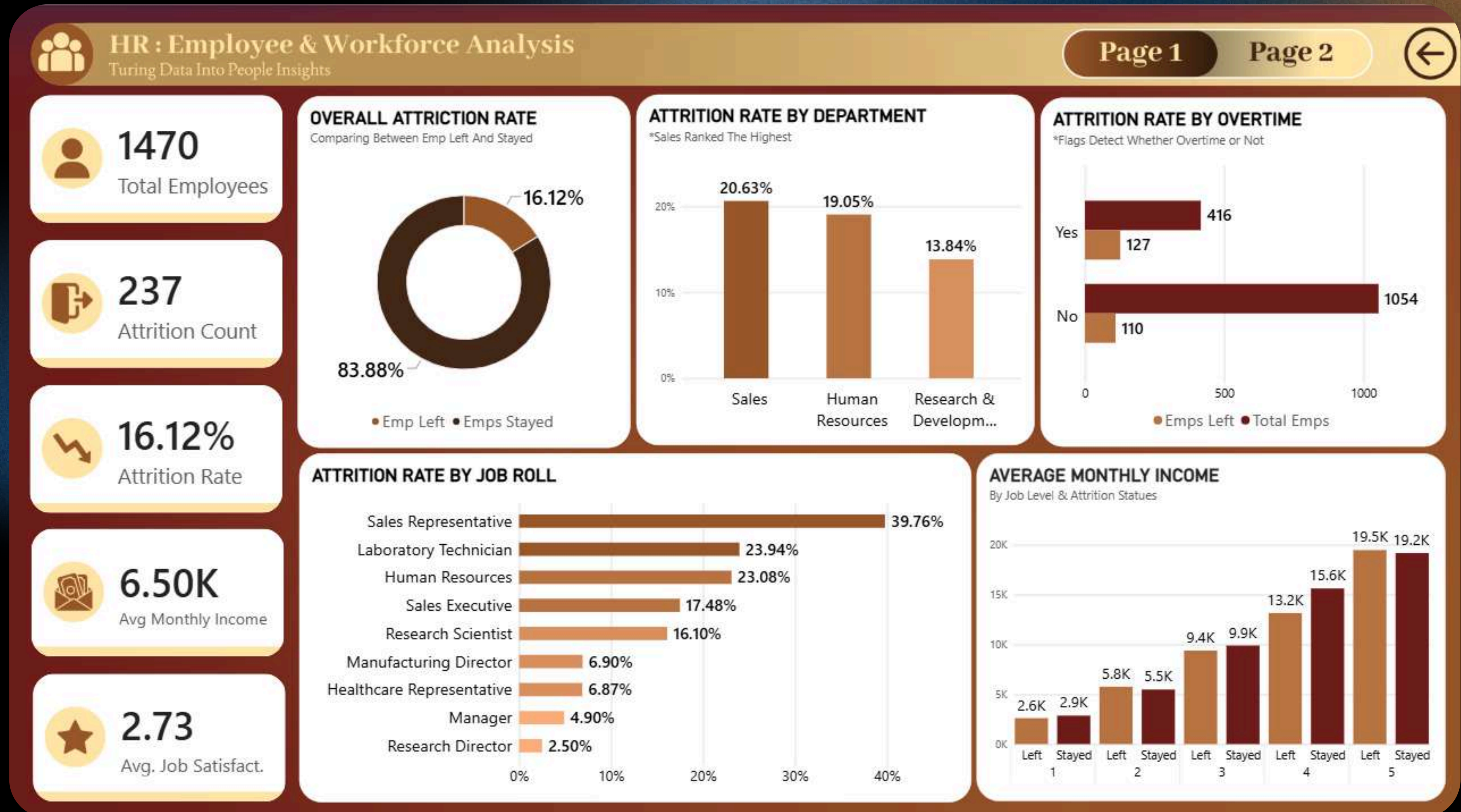
05 DASHBOARDING & REPORTS

Employee Attrition & Workforce Analytics Dashboard

START EXPLORING

WORKFORCE INSIGHTS - HR ANALYTICS

05 DASHBOARDING & REPORTS



05 DASHBOARDING & REPORTS



HR : Employee & Workforce Analysis

Turing Data Into People Insights

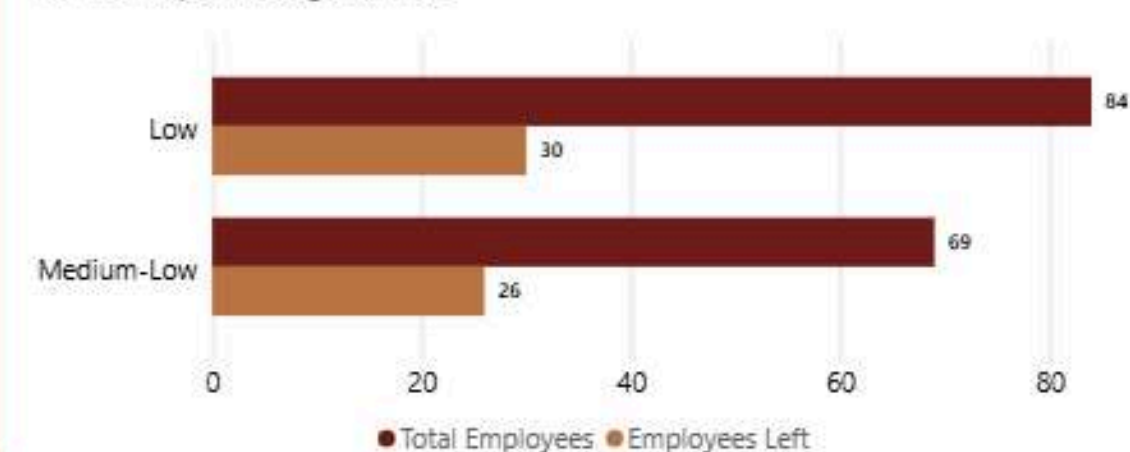
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ATTRITION RATE BY LOW JOB SATISFACTION LEVELS

*For The People Working Overtime

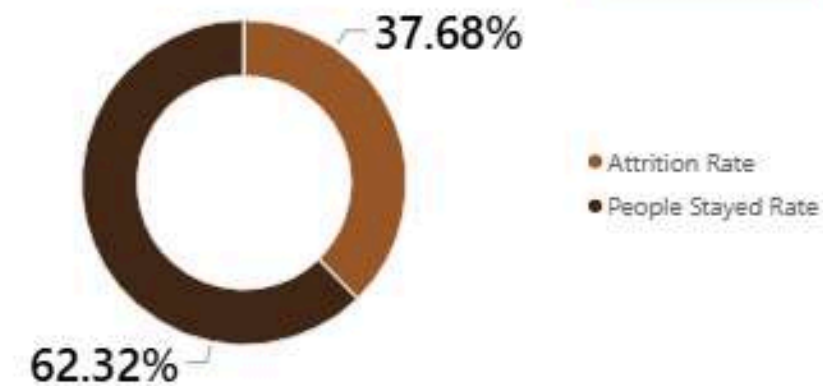


DRILL THROUGH BY JOB SATISFACTION

Attrition Rate and People Stayed Rate

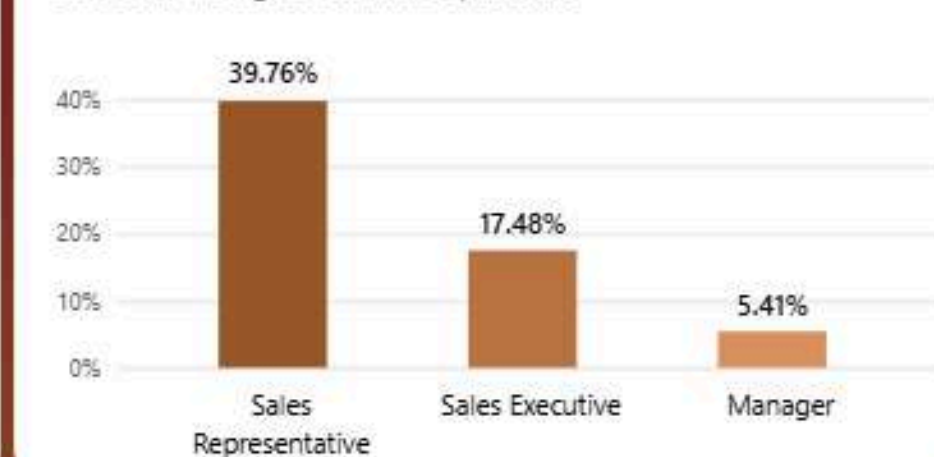
Low

Medium-Low



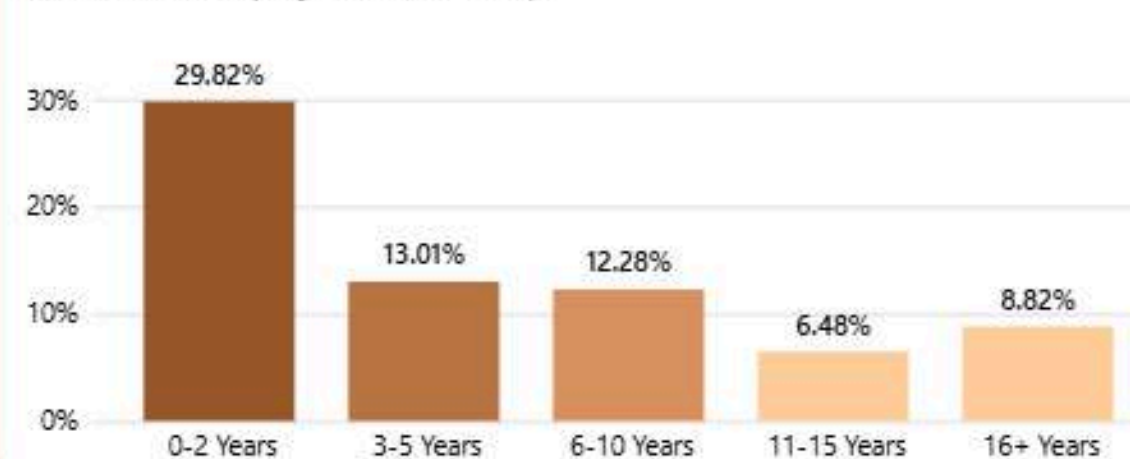
ATTRITION RATE BY HIGHEST DEPARTMENT

*Sales Was The highest attrition department



ATTRITION RATE BY COMPANY TENURE GROUP (%)

Bucket 'YearAtCompany' Into Tenure Groups



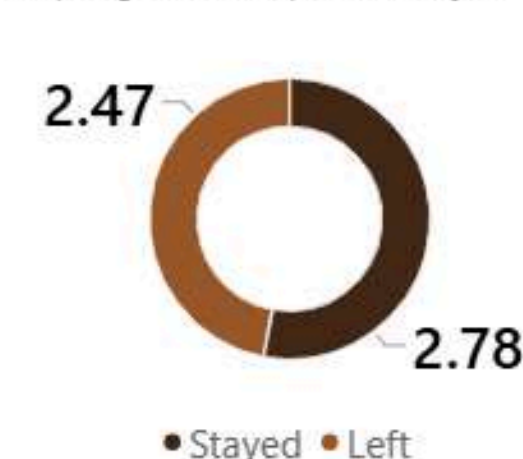
AVG MONTHLY INCOME (\$)

Comparing between Emps Left VS Stayed



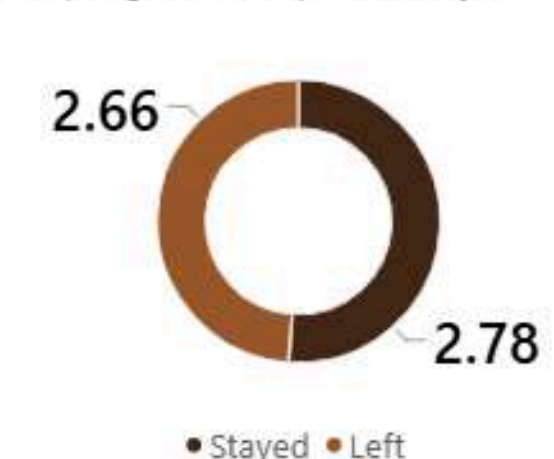
AVG JOB SATISFACTION

Comparing between Emps Left VS Stayed



AVG WORK-LIFE BALANCE

Comparing between Emps Left VS Stayed



Key Insights

- Reduce excessive overtime, especially in the Sales department.
- Prioritize retention strategies for Sales Representatives, the single highest-risk role.
- Increase employee engagement and job satisfaction, particularly for employees combining low satisfaction with overtime.
- Review compensation for lower-income employee groups, since leavers earn noticeably less on average.
- Strengthen onboarding, mentoring, and early check-ins to support employees through their first two years.

THANK YOU!